1MA188, 2025-05-26, Solutions

$\boxed{\text{Pr 1}}$

(a) $[T \circ T] = [T][T] = \begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix}\begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix} = \begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix}$

(b) Eigenvalues : need to solve

$$\det([T] - \lambda I) = \det\begin{pmatrix} 1/5 - \lambda & -2/5 \\ -2/5 & 4/5 - \lambda \end{pmatrix} = 0$$

$$\lambda^2 - \lambda = 0$$

$$\lambda(\lambda - 1) = 0 \quad \Rightarrow \quad \lambda = 0 \, , \, \lambda = 1$$

Eigenvectors for $\lambda = 0$:

$$\begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} v_1 - 2v_2 = 0 \\ -2v_1 + 4v_2 = 0 \end{cases}$$

$$\Rightarrow \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} t \quad \text{for any} \quad t \neq 0$$

Eigenvectors for $\lambda = 1$:

$$\begin{pmatrix} -4/5 & -2/5 \\ -2/5 & -1/5 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -4v_1 - 2v_2 = 0 \\ -2v_1 - v_2 = 0 \end{cases}$$

$$\Rightarrow \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} t \quad \text{for any} \quad t \neq 0$$

(c) Since $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ satisfies $T\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$, we can take it as a directional vector

So $\begin{cases} x = -t \\ y = 2t \end{cases}$ is our line

(can also be written as $y = -2x$)

Pr 2

(a)

$$\det\begin{pmatrix} x & x & x & x \\ 1 & x & x & x \\ 2 & 2 & x & x \\ 0 & 2 & -1 & x \end{pmatrix} = x \det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & x & x & x \\ 2 & 2 & x & x \\ 0 & 2 & -1 & x \end{pmatrix}\begin{array}{l} \\ -\text{ I row} \\ -2\times\text{ I row} \\ \\ \end{array}$$

$$= x \det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & x-1 & x-1 & x-1 \\ 0 & 0 & x-2 & x-2 \\ 0 & 2 & -1 & x \end{pmatrix} = x(x-1)\cdot\det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & x-2 & x-2 \\ 0 & 2 & -1 & x \end{pmatrix}\begin{array}{l} \\ \\ \\ -2\times\text{II} \end{array}$$

$$= x(x-1)\det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & x-2 & x-2 \\ 0 & 0 & -3 & x-2 \end{pmatrix} = x(x-1)(x-2)\det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -3 & x-2 \end{pmatrix}\begin{array}{l} \\ \\ \\ +3\times\text{III} \end{array}$$

$$= x(x-1)(x-2)\det\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & x+1 \end{pmatrix} = x(x-1)(x-2)(x+1)$$

(b) A is invertible $\iff \det A \neq 0 \iff x \neq 0, 1, 2, -1$

Pr 3

(a) Use polar coordinates:

$$\lim_{\substack{r \to 0 \\ \text{uniformly} \\ \text{in } \theta}} \frac{r \sin\theta \, \sin(r^2)}{r^2}$$

$$0 \le \left| \frac{r \sin\theta \, \sin(r^2)}{r^2} \right| \le \left| \frac{\sin(r^2)}{r} \right|$$

$0 \to 0$; $\left| \frac{\sin(r^2)}{r} \right| \to 0$ by L'Hospital Rule

$$\lim_{r \to 0} \frac{\sin(r^2)}{r} = \lim_{r \to 0} \frac{\cos(r^2) \cdot 2r}{1} = 0$$

$\left| \frac{r \sin\theta \, \sin(r^2)}{r^2} \right| \to 0$ by Squeeze theorem

Answer: limit exists and is 0

(b) Take path $x = 0, y = 0, z = t$

$$\frac{0 + 0 + 0}{t^2} \to 0 \quad \text{as } t \to 0$$

Take path $x = 0, \; y = z = t$

$$\frac{0 + 0 + t^2}{0^2 + t^2 + t^2} = \frac{1}{2} \to \frac{1}{2} \quad \text{as } t \to 0$$

So limit doesn't exist

(more generally, take line $\begin{cases} x = k_1 t \\ y = k_2 t \\ z = k_3 t \end{cases}$:

$$\frac{k_1^2 k_2^2 t^4 + k_1^2 k_3 t^3 + k_2 k_3 t^2}{(k_1^2 + k_2^2 + k_3^2) t^2} \longrightarrow \frac{k_2 k_3}{k_1^2 + k_2^2 + k_3^2} \quad \text{as } t \to 0$$

Clearly this depends on the choice of k_1, k_2, k_3, so limit doesn't exist

Pr 4

(a) $f_x = 2x$

$f_y = 2y$

At $(0,0,0)$: $z = 0 + 0(x-0) + 0(y-0)$ - tangent

So $z = 0$

(b) Let $g(x,y) = -(x+1)^2 - y^2 + 1$

$g_x = -2(x+1)$

$g_y = -2y$

At $(0,0,0)$, tangent plane is

$$z = 0 - 2(x-0) + 0(y-0)$$

So $z = -2x$

(c) It's intersection of $z=0$ and $z=-2x$

$z = 0 \Rightarrow x = -\frac{z}{2} = 0$

So tangent line is $\begin{cases} x = 0 \\ y = t \\ z = 0 \end{cases}$, i.e. y-axis

(d) To find projection onto the xy-plane, we eliminate

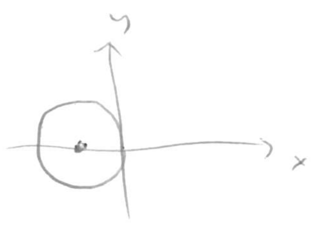
z-variable: $\begin{cases} z = x^2 + y^2 \\ z = -(x+1)^2 - y^2 + 1 \end{cases}$

$\Rightarrow x^2 + y^2 = -(x+1)^2 - y^2 + 1$

$2x^2 + 2x + 2y^2 = 0$

$2(x+\frac{1}{2})^2 + 2y^2 = \frac{1}{2}$

$(x+\frac{1}{2})^2 + y^2 = \frac{1}{4}$ - circle centered at $(-\frac{1}{2}, 0)$ of radius $\frac{1}{2}$

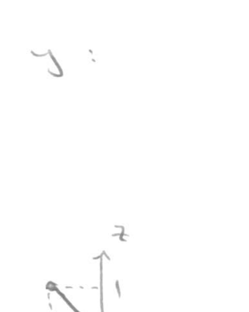


(e) To find projection onto xz-plane, we eliminate y:

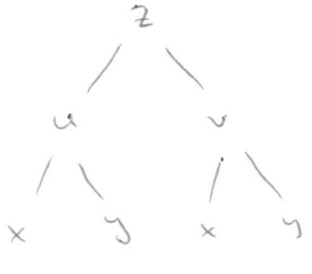
$\begin{cases} z = x^2 + y^2 \\ z = -(x+1)^2 - y^2 + 1 \end{cases} \Rightarrow 2z = x^2 - (x+1)^2 + 1$

$2z = -2x$, i.e. $z = -x$. That's a line

From (d) it is clear that $-1 \le x \le 0$, so interval:



Pr 5



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} = \frac{\partial z}{\partial u}\cdot 3x^2 + \frac{\partial z}{\partial v}\cdot 1$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y} = \frac{\partial z}{\partial u}\cdot(-1)$$

PDE becomes

$$\cancel{3x^2\frac{\partial z}{\partial u}} + \frac{\partial z}{\partial v} + \cancel{3x^2\frac{\partial z}{\partial u}\cdot(-1)} = xy$$

$$\frac{\partial z}{\partial v} = xy$$

Need to express x and y in terms of u, v:

$$\begin{cases} x = v \\ y = x^3 - u = v^3 - u \end{cases}$$

So

$$\frac{\partial z}{\partial v} = v(v^3-u) = v^4 - uv$$

Integrate with respect to v:

$$z = \frac{v^5}{5} - u\frac{v^2}{2} + g(u)$$ for any differentiable function $g(t)$

Answer: $z = \frac{1}{5}x^5 - \frac{1}{2}x^2(x^3-y) + g(x^3-y)$

Pr. 6

(a) $\begin{cases} f_x = 2x + 2xy = 0 & \Rightarrow \quad x(1+y) = 0 \\ f_y = 4y + x^2 = 0 \end{cases}$

Case 1 : $x = 0 \Rightarrow y = 0$

Case 2 : $1 + y = 0 \Rightarrow y = -1 \Rightarrow x^2 = -4y = 4$

$\Rightarrow x = \pm 2$

So $(0,0)$, $(2,-1)$, $(-2,-1)$ are critical

(b) $f_{xx} = 2 + 2y$

$f_{xy} = 2x$

$f_{yy} = 4$

$H = \begin{pmatrix} 2+2y & 2x \\ 2x & 4 \end{pmatrix}$ - Hessian

At $(0,0)$: $H = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ positive definite

$\Rightarrow (0,0)$ is local min

(alternatively : $Q = 2k^2 + 4h^2 > 0$)

At $(2,-1)$: $H = \begin{pmatrix} 0 & 4 \\ 4 & 4 \end{pmatrix}$ $D_1 = 0$, $D_2 = -16 \neq 0$ - indefinite

so saddle point

(alternatively : $Q = 8kh + 4h^2 = 4(h+k)^2 - 4k^2$ - indefinite)

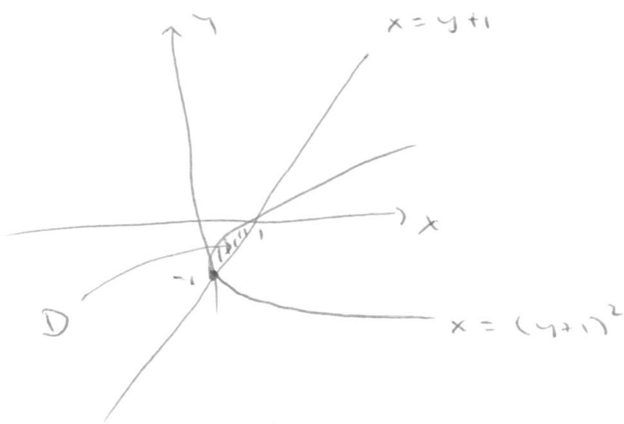
At $(-2,-1)$: $H = \begin{pmatrix} 0 & -4 \\ -4 & 4 \end{pmatrix}$ $D_1 = 0$, $D_2 = -16 \neq 0$ - indefinite

(or : $Q = -8kh + 4h^2 = 4(h-k)^2 - 4k^2$ - indefinite)

so $(-2,-1)$ is also a saddle point

Pr 7

$$\begin{cases} y = x - 1 \\ x = (y+1)^2 \end{cases}$$



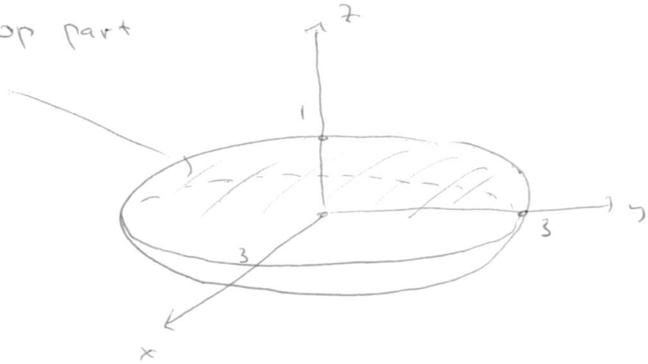
$$\text{Area}(D) = \iint_D 1 \, dx\, dy = \int_{-1}^{0} dy \int_{(y+1)^2}^{y+1} dx$$

$$= \int_{-1}^{0} \left(y+1 - (y+1)^2 \right) dy = \int_{-1}^{0} (-y^2 - y)\, dy$$

$$= \left(-\frac{y^3}{3} - \frac{y^2}{2} \right) \Big]_{y=-1}^{y=0} = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$$

Pr 8

E is top part



Cylindrical coordinates:

E is $0 \le \theta \le 2\pi$

$0 \le r \le 3$

$r^2 + 9z^2 \le 9$ i.e. $|z| \le \sqrt{1 - \frac{r^2}{9}}$

$z \ge 0$

$$\iiint_E (r\cos\theta + r\sin\theta + z)\, r\, dr\, d\theta\, dz$$

$$= \int_0^{2\pi} d\theta \int_0^3 r\, dr \int_0^{\sqrt{1-\frac{r^2}{9}}} (\underbrace{r\cos\theta}_{\text{will give zero because } \int_0^{2\pi}\cos\theta\, d\theta = 0} + \underbrace{r\sin\theta}_{\text{same reason}} + z)\, dz$$

$$= \int_0^{2\pi} d\theta \int_0^3 r\, dr \int_0^{\sqrt{1-\frac{r^2}{9}}} z\, dz = 2\pi \int_0^3 \left[\frac{z^2}{2}\right]_{z=0}^{z=\sqrt{1-\frac{r^2}{9}}} r\, dr$$

$$= 2\pi \int_0^3 \frac{\left(1-\frac{r^2}{9}\right) r}{2}\, dr = \pi \left(\frac{r^2}{2} - \frac{1}{9}\cdot\frac{r^4}{4}\right)\Big|_0^3 =$$

$$= \pi\left(\frac{9}{2} - \frac{1}{9}\cdot\frac{81}{4}\right) = \frac{9\pi}{4}$$